

Project Problem Proposal and Approval Form

Field	Team response
Team number	17
Working project title	University of Botswana Service Queue And Appointment Management System
Members and student IDs	Phenyo Morapedi 20210512 Boikhutso Sugape 201901809 Aone Ropane 202402953 Phenyo Joseph Mpala 202401164
Repository URL	https://github.com/CSI473GroupProject/University-Servive-Queue-and-Appointment-Management-System.git
Proposal tag/commit	
Date submitted	11 August 2026

Originality declaration: This proposal is based on our own investigated problem. It is not the University Service Hub lecturer case.

1. Problem statement (150-250 words)

Students at the University of Botswana may experience difficulties when accessing student registration services, particularly during busy registration periods. The current process can result in students spending long periods waiting to receive assistance, while the handling of students in the queue may not always be efficient. This can create congestion around registration service points and make it difficult for students to know when they will be served.

The problem affects students who need registration-related assistance as well as staff members responsible for providing these services. Students may have to spend significant amounts of time waiting, which can interfere with classes, studying and other university activities. Staff may also experience difficulties managing large numbers of students requesting assistance at the same time.

The main goal of the proposed project is to reduce student waiting times by improving how students join and move through the registration service queue. The system will also provide an appointment mechanism that allows students to select available service times where appropriate. By organising students according to their queue or appointment status, the system aims to make the registration service process more orderly, predictable and efficient.

2. Evidence of the problem

Evidence/source	Date/origin	What it shows
Student observations/experiences	27 July 2026	Students experience long waiting times when accessing registration services.
Student interviews/questionnaires	04 August 2026	Students report difficulties associated with waiting and accessing registration assistance.
Observation of Registration service process	28 July 2026	Large numbers of students may need assistance during busy registration periods, creating pressure on the service queue

3. Goal and objectives

Item	Team response
To investigate the causes and effects of long queues and waiting times experienced by University of Botswana students when accessing university services	To improve the efficiency and accessibility of student services at the University of Botswana by reducing unnecessary waiting times and improving queue and appointment management.
To identify the requirements of students and university service staff for an effective queue and appointment management process.	To provide an appointment facility that allows students to select available registration service times and receive confirmation of their appointments.
To design and evaluate a prototype that enables students to manage appointments and monitor their position in service queues.	To provide registration staff with a system for monitoring queues and appointments so that students can be served in an organised and efficient manner.

4. Stakeholders

Stakeholder/role	Need	Concern	Evidence/access
Students	They need convenient way to access way to access university services without spending excessive time waiting in physical lines.	Long waiting periods and uncertainty about when they will be served.	Student interviews/questionnaires and system access.
Registration staff	Need to manage students waiting for registration assistance efficiently.	Difficulty managing large numbers of students during busy periods.	Staff interviews/observations and staff system access.
Registration/service administrators	Need to monitor appointments, queues and service activity.	Accurate information and effective control of service availability.	Administrative access to the system.
Project team	Need accurate requirements and feedback for designing the system	Ensuring the system addresses the actual registration-service problem.	Interviews, observations and testing.

5. Scope, assumptions and constraints

In scope	Out of scope
Student registration service queue management.	Processing student tuition or fee payments.
Student registration service appointments	Academic advising outside the registration service.

Student registration service selection.	Replacing the University's complete student information system.
Joining and monitoring a virtual queue.	Managing unrelated university services.

Assumptions	Constraints
Students have access to a phone, computer or other internet-enabled device.	The development team has limited time and resources.
Registration staff will have access to the system.	The system must protect student information and avoid unnecessary collection of personal data.

6. Proposed core workflow / vertical slice

1. A student logs into the Student Registration Service Queue and Appointment Management System and selects the registration service they require.
2. The student chooses either to join the virtual queue or select an available appointment time.
3. If the student joins the queue, the system assigns a queue number and displays the student's current queue status.
4. If the student selects an appointment, the system checks whether the requested time is available and confirms the appointment if a slot is available.
5. The system updates the student's status as the queue progresses or the appointment approaches.
6. Registration staff view the active queue and appointments and call the next eligible student according to the applicable service rule.
7. The student's status changes to served/completed after the registration service has been provided.
8. If an appointment slot is unavailable or the queue is full, the system informs the student and provides an appropriate alternative, such as another available appointment or queue option.

Important rule: A student should not be assigned to an unavailable appointment slot, and the system should maintain an accurate queue position/status.

State/result created: A queue record or appointment record is created and its status changes as the student moves through the service process.

Failure/exception: The system must handle situations where an appointment slot is already full, a student cancels an appointment, or a student does not attend the scheduled appointment.

7. Design-suitability check

Check	Yes/No	Short evidence
Clear stakeholders and a genuine current problem	yes	Students and registration staff are directly affected by waiting times and queue-management difficulties.
At least one state-sensitive workflow or entity	yes	Appointment and queue records can move through states such as booked, waiting, called, served, cancelled or missed.
Important business/validation/security rule	yes	The system must prevent double-booking and ensure that only available appointment slots can be selected.

Relevant quality trade-offs	yes	The system must balance usability, performance, reliability and protection of student information.
Relevant failure/exception condition	yes	The system must handle unavailable appointment slots, cancellations, missed appointments and queue changes.
Feasible semester-sized vertical slice	yes	A basic queue and appointment workflow can be implemented as a focused semester project.
Synthetic/anonymised data is sufficient	yes	Development and testing can be conducted using synthetic student and appointment information.
Not the University Service Hub or a renamed copy	yes	The proposed system specifically focuses on student registration service queues and appointments rather than acting as a general university service hub.

8. Data, ethics, safety and dependency risks

Risk/concern	How the team will control it
Exposure of student personal information	Use only the minimum information required and use synthetic/anonymised data during development and testing.
Unauthorised access	Implement authentication and role-based access for students, staff and administrators.
Double booking	Prevent two students from receiving the same appointment slot.

9. Lecturer decision

Decision	Select one
Approved	☐
Approved with minor conditions	☐
Revise and resubmit	☐

Field	Complete
Conditions/comments	
Lecturer	Dr Cleverence Kombe
Decision date / resubmission date if applicable	